

WASP-36b

R-1054

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_170403 · created 2026-07-30T10:31:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457846.6758 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.048
Transit depth (Rp/Rs)^2	1.03 +/- 0.97 [%]
Orbital Inclination (inc)	79.12 +/- 2.86
Airmass coefficient 1 (a1)	7.05 +/- 4.02
Airmass coefficient 2 (a2)	-1.3 +/- 0.76
Scatter in the residuals of the lightcurve fit is	63.42 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.5
Transit Duration (day)	0.026 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.101 ± 0.048 vs 0.1368 (deviation 0.75σ)
Duration fit vs published	0.026 d vs 0.0758 d (deviation 1.66σ)
Mid-time O–C	-0.8 min
Depth SNR	2.1
Residual scatter	63.42 %

Light curve

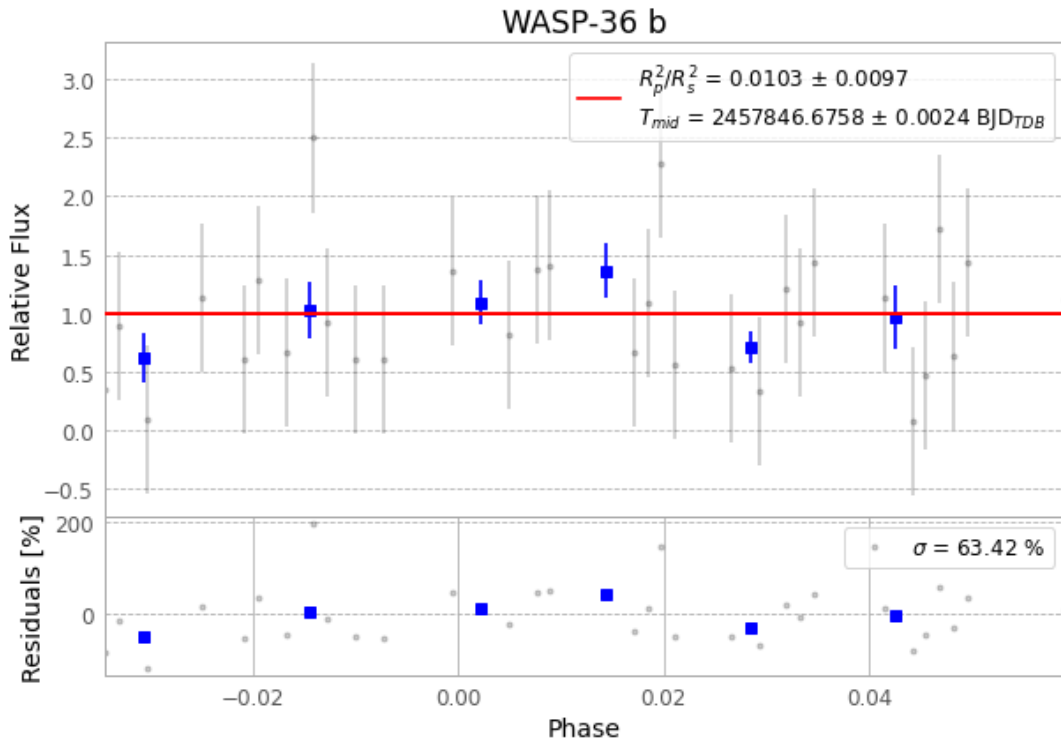


Figure 1 — FinalLightCurve_WASP-36 b_2017-04-02.png (SHA-256 109b95a22fba31f9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [94, 430], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9fc474727da029dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-36170403023914.FITS → WASP-36170403062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-170403.log (122113401b3a...), FinalLightCurve_WASP-36 b_2017-04-02.png (109b95a22fba...), FinalLightCurve_WASP-36 b_2017-04-02.csv (a73e7551a534...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 10:31Z.